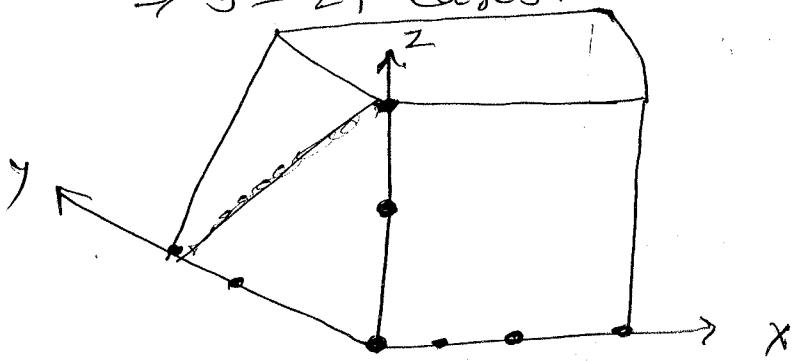
ex: 3 inputs. x, y, z.

⇒ 3 distinct values → yes, No, May be.

⇒ $3^3 = 27$ Cases.



Now based on 1 value, we keep deciding & solving.

Introduction to Testing

Testing is the process of exercising a program with specific intent of finding errors prior to delivery to the end users.

(or)

A process of executing a program with an intent of finding an error.

The main objective of testing is to prove that the s/w product meets a set of pre-established acceptance criteria under a prescribed set of environment circumstances.

A strategic approach to S/w Testing:

General characteristics:

→ To perform effective testing, a S/w team should Conduct technical review.

→ Testing begins at Component level & continues till end.

→ Different testing techniques are appropriate at different points in time.

→ Testing is done by developer of S/w & for large projects by an independent test group.

→ Testing & debugging are different activities.

→ Verification & validation are two important aspects of S/w testing process.

Verification: " "Are we implementing the software right?" " Does the product meet its specifications

- The set of activities that ensure that S/w correctly implements a specific function or algorithm.

Validation:

"Are we implemented the right software?"

"Does the product perform as desired?.

→ The set of activities that ensure that the s/w has been built is traceable to customer requirements.

~~as per the goal of Customer requirements.~~

Verification Example:

Developer developed:

→ Chatting functionality then verify it.

→ Audio call functionality then verify it

→ Share photos functionality " " ".

→ Video call functionality " " ".

Validation example:

Tester Validate Complete developed product at the same time.

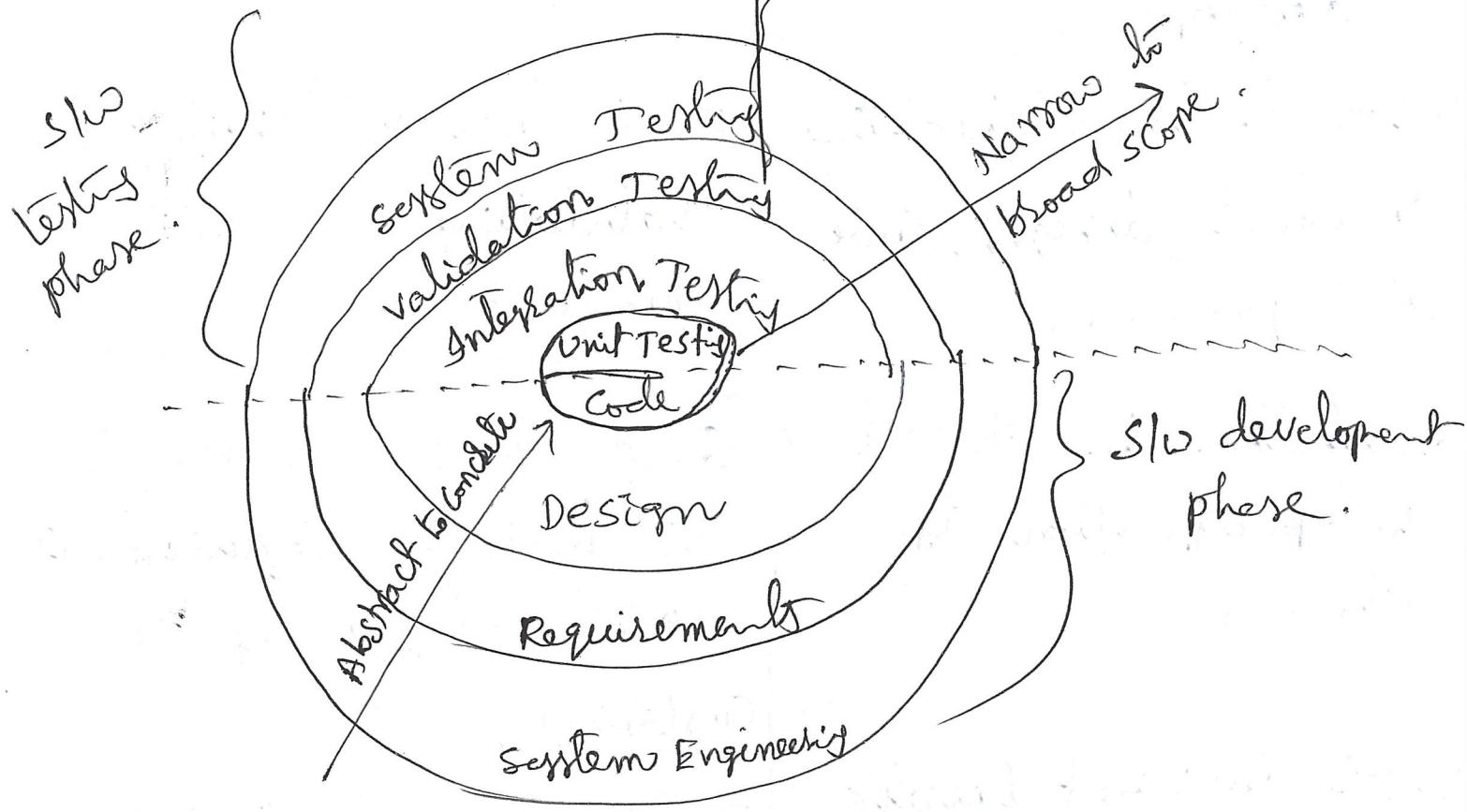
→ Validate chatting, Audio/video call, share photos functionality at a same time.

→

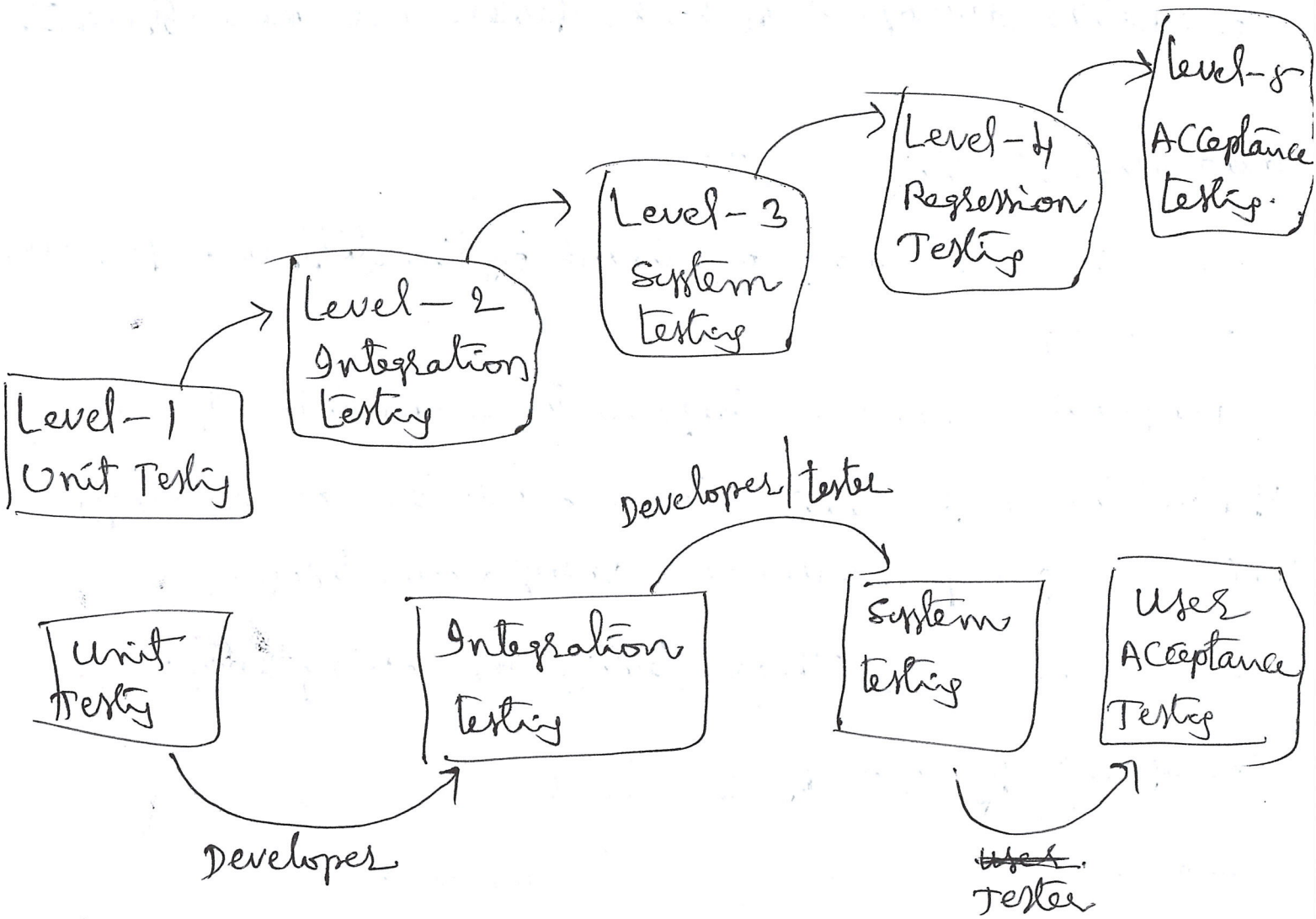
Verification	Validation
1. It means "Are we implementing the S/w right?"	1. It means "Are we implemented the right S/w?"
2. It comes before Validation	2. It comes after Verification
3. It is known as static testing.	3. Dynamic testing.
4. It is executed by Quality assurance team or developer.	4. It is executed by the Testing team.
5. It includes checking documents, design, code and program.	5. It includes testing of validating the actual product.
6. Whether the S/w confirms to specification is checked.	6. It checks whether the S/w meets the requirements and expectations of a customers.
7. It does not involve executing the code.	7. It always involve executing the code.
8. It finds bugs early in the development cycle.	8. It can find bugs that the verification process can not catch.

9. It includes different methods like Inspections, Reviews & walkthroughs.	9. It includes testing like functional, system, Integration & user acceptance testing.
10. Developer can verify that the i/ps follow the o/ps or not.	10. Testers can validate that the user accepts the product or not.

Test strategies for Conventional S/w:



Levels of Testing



Unit testing :-

→ focuses on testing each module/component/independent units.

→ A unit may be an individual function, method, procedure, object in a program

→ It is also called as component testing.

→ It uses white box testing techniques

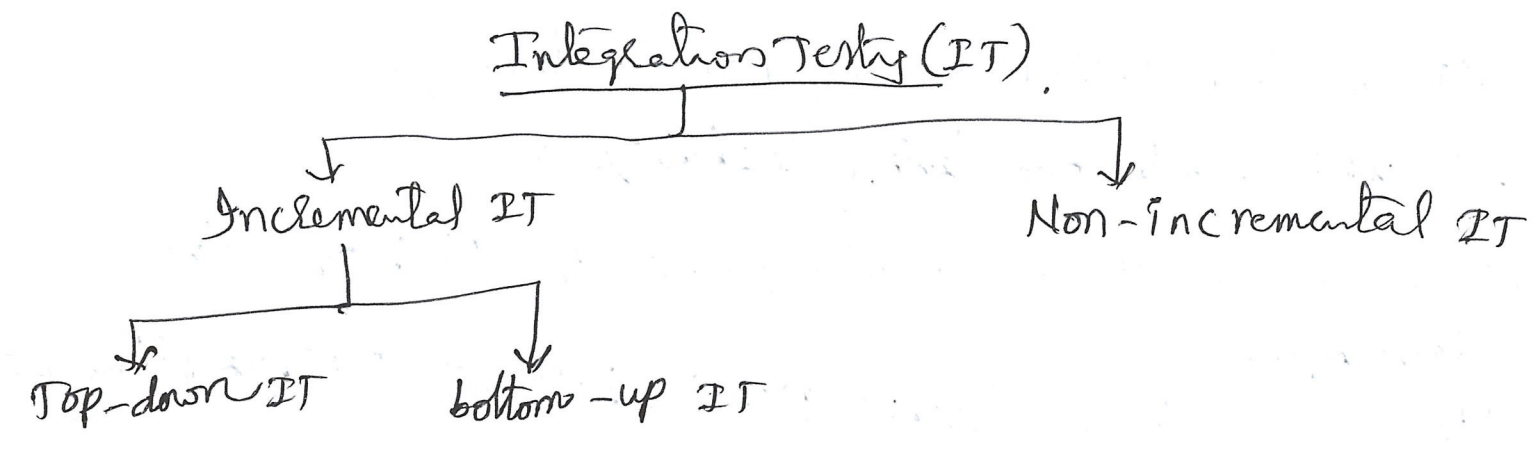
→ It is done during coding phase of SDLC by S/w developers

→ Unit testing tools: JUnit, NUnit, PHPUnit, EMMA & Jmockit

→ It ensures that each unit of code meets the requirements, improving the over all quality of the s/w.
→ It enables developers to work faster and more efficiently.

Integration Testing :- (IT)

→ All modules may be coded by the different programmers or developers.
→ Integrating various components or modules of an application & testing their behavior as a single unit or group called as Integration testing.
→ Also known as Thread testing or String testing.
→ The goal of IT is to check Correctness & Communication among all modules.
→ Integration Testing tools : Selenium, PyTest, Junit, Jasmine, steam, Mockito, etc.
→ It can test h/w's Compatibility with s/w.



① Incremental IT: Modules are added in ascending order one by one or according to need. selected modules must be logically related.

Two types:

(a) Top-Down: It starts by testing top-most modules and moving down to the lowest set of modules one-by-one & check control flow of the system.

(b) Bottom-up IT: starts with testing the lowest units of the application and moving up one-by-one. Testing takes place from the bottoms of the control flow to upwards.

② Non Incremental testing / Big Bang testing:

once all the modules are developed and tested individually, they are integrated once and tested together at once. No parent & child concept.

System testing : (ST)

→ ST validates the complete and fully integrated s/w product.

→ The purpose of ST is to evaluate the end-to-end system specifications

→ It is a type of Black box testing perform by Quality Assurance Team in Testing phase.

System Testing tools :- Selenium, LoadRunner, Jmeter, Microsoft Test manager, SoapUI.

→ The choice of tools depends on various factors like. technology used, size of the project, Budget & Testing requirements.

→ Importance of system testing : Error reduction, Cost Savings, Security, Improved product quality, s/w performance, Customer satisfaction :

Types of ST :-

@ Performance testing : measures speed, load time, stability, reliability, response time of the system under various conditions.

(b) load testing: It determines how systems performs under a real-life extreme load (Throughput, No. of users)

(c) usability testing:

It checks whether the systems is easy to use and functional for end users.

(d) Regression testing: It ensures that any changes done during the development process have not introduced a new defects or bugs.

(e) Migration testing:- It ensures that if system needs to be modified in new infrastructure so it should be modified without any issue.

(f) functional testing:-

Tester finds out any missing function in the system & make list of it. It can be added during the functional testing.

(g) Recovery testing: It ensures that systems is capable of recovering from certain system errors, crashes and failures.

(h) stress testing: It ensures that robustness of the system under various loads & capacities.

① S/w & H/w testing:-

It checks the H/w & S/w compatibility and interaction

Acceptance Testing:

It is a final level of testing before making the system available for actual use.

- It is a type of Black box testing & performed by end user or client.
- It should be performed on live environment or in real time scenarios.
- also called as user Acceptance Testing (UAT), Functional acceptance testing (FAT) and, End user Testing (EAT) or Red Box Testing (RAT).

Acceptance testing tools: Fitness tools, Watir etc.

- It helps to deliver the end product without any bugs.

Types: UAT, BAT, RAT

② User Acceptance Testing (UAT):-

It checks whether the s/w product is working as per the requirements & correctly before moving into the production environment.

- This testing is performed in a different production like set-up environment.

Business Acceptance testing (BAT) :- It checks whether the S/w product is able to meet the business requirements and operational need.

- mainly focuses on business risk & financial factors.

Regulations Acceptance Testing (RAT) :

It checks whether the S/w product developed with rules & regulations of the country/region where it is getting released.

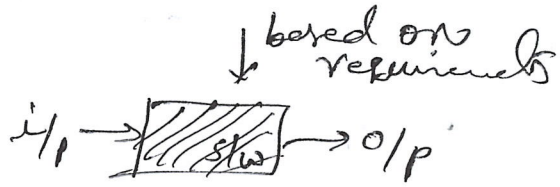
Black box testing

→ BBT is done without the internal knowledge of the software.

→ This method is applied at every level of s/w testing. i.e, unit testing, integration testing, system testing, Acceptance testing.

→ It doesn't check the internal code of the s/w.

→ It checks the behaviours, hence it is also called the behavioural testing.



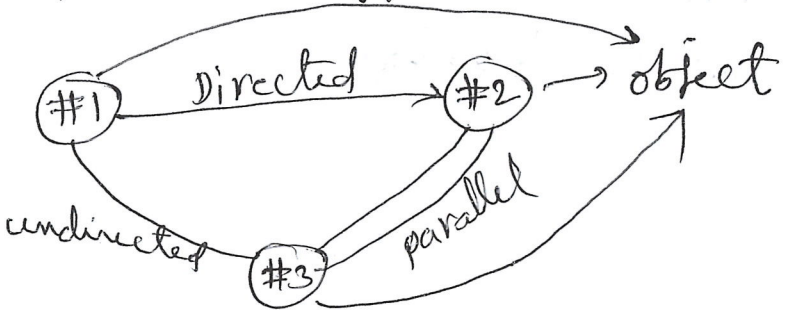
BBT.

→ Tests are based on requirements and functionality.

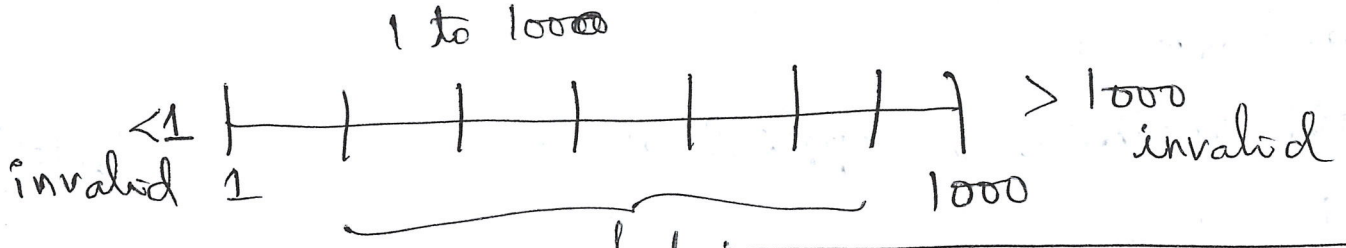
Methods:

① Graph based method:

This technique of BBT involves a graph drawing that dipicts the link b/w 'cause (i/p ⇐)' & the effects (o/p), with trigges the effects



② Equivalence Partitioning : It divides the i/p domain into multiple partitions.



* i/p values that provide to system are divided into different classes or groups based on its similarity in the outcome.
* Instead of using each & every i/p value, use any one value from the group to test outcome.

Ex:

① Value b/w 1 to 99

1 – 99 → valid

<1 } invalid
>99 }

② value = 19

19 → valid

<19 } invalid
>19 }

AGE: [ ] * Accepts values from 18 to 60

Equivalence class partitioning		
Invalid	Valid	Invalid
<=17	18–60	>=61

③ in a set { 1, 2, 3, 4, 5 }

If value is in set → Valid.

" " " not in the set → invalid

④ Boolean:

True → valid

False → invalid.

Black box testing techniques

③ Decision Table Testing :-

→ various i/p combinations & their respective system behaviours are captured in a tabular form.

→ check logical relationship b/w two and more than two i/ps.

ex: E-mail account :-

Email (Condition-1)	T	T	F	F
password (Condition-2)	T	F	T	F
Expected Result (Action)	Account page	Incorrect password	Incorrect email	Incorrect email.

④ Error Guessing :

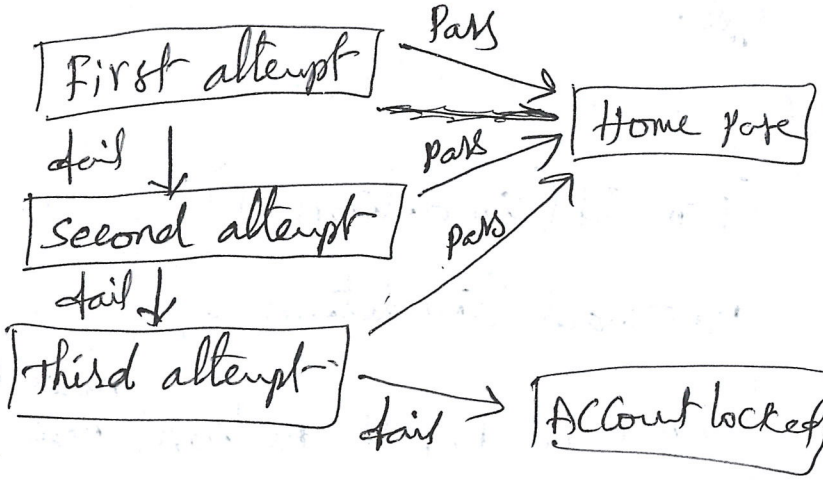
- It is based on the experience of the tester, where tester uses experience to guess the problematic areas of the s/w.

ex: Divide by zero, Handling null values in the text fields, Accepting the submit button without any value, file upload without attachment, file upload with less than or more than the limit size.

⑤ State transition testing:

→ It is used to capture the behavior of the S/W application when different input values are given to the same function.

→ Applies to those types of functions that provide specific no. of attempts to access applications.



***Disadvantages of BBT:-**

- It does not reveal the errors in the control structure.
- Without clear programing knowledge, test cases are difficult to implement.

⑥ All pairs testing technique

- It is used to test all possible discrete combinations of values.
- This combinational method is used for testing the application that uses Check box i/p, radio button i/p, list box, text box etc.

Advantages of BBT:

- Tester does not need to have programing skills / functional knowledge.
- It is efficient for implementing the tests in the larger system.
- Tests are executed from the user's/client's point of view.

(7)

(#) Boundary value Analysis: (BVA)

(2)

→ More application errors occurs at boundaries

* Boundary values are those that contain the upper & lower limit of a variable.

→ It is used to identify errors at boundaries rather than finding those errors exist in the centre of the i/p domain.



→ It tests, while entering boundary values, whether the s/w is producing correct o/p or not.

→ While writing program with loop, we always think twice loop should start from i=0 or i=1 and conditions > or >= and < or <=

→ Hence to resolve this, we always check values at both boundaries.

-1 0 | 1 2 ..... 10 | 11 12

AGE: [ ] 18-56		
Boundary value Analysis		
Invalid min-1	valid min, +min, -max, max	invalid max+1
17	18, 19, 55, 56	57

(8)

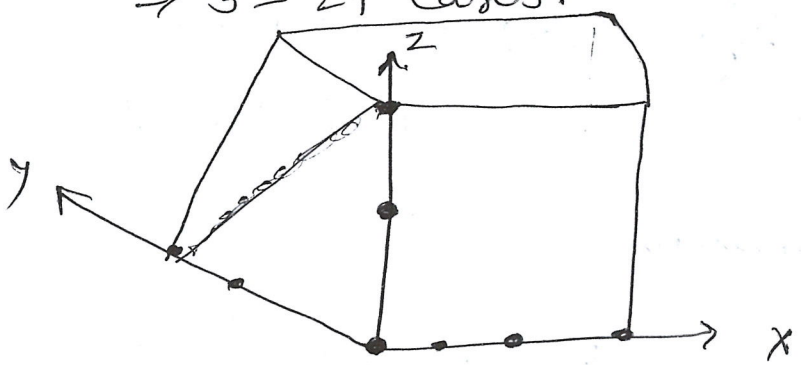
(A) Orthogonal array testing:

The problem in which the i/p domain is small but it is large enough to accomodate the exhaustive testing.

ex: 3 inputs x, y, z.

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2. Difference between **White Box Testing** and **Black Box Testing**:

	White Box Testing	**Black Box Testing**
1	The internal structure & working of the software did not known to the tester. Focus on Input & Output of Product.	The internal structure & working of the software is known to the tester.
2	Is also known as Functional Testing, Data-driven testing, Closed-box testing & Behavioral Testing.	It is also known as Structural testing, Glass box testing, Code-based testing, and Transparent testing.
3	Less programming knowledge is required.	Requirement of complete programming knowledge.
4	It is done at higher levels of testing that are system and acceptance testing.	It is done at lower levels of testing that are unit testing and integration testing.
5	Mainly performed by Software Testers.	Mainly performed by Developers.
6	Less time-consuming	More time-consuming
7	The base of this testing is external expectations.	The base of this testing is coding which is responsible for internal working.
8	Its main objective is to fulfill customer need.	Its main objective is to check the code quality.
9	Defects are identified once the code is ready.	Defects are identified at early stage
10	Types of black-box testing:	Types of white box testing:

	Functional & Non-Functional Testing and Regression testing.	Path testing, Loop testing and Condition testing.
11	Can be done by trial and error ways and methods.	Data domains along with inner or internal boundaries can be better tested.
12	Example: Search something on Google by using keywords	Example: By input to check and verify loops

3. Difference between Alpha and Beta Testing:

	Alpha Testing	**Beta Testing**
1	It performed by a team of highly skilled testers who are usually the internal employee of the organization.	It performed by clients or end-users in client location or in real-time environment.
2	It involves both white box & black-box techniques.	It uses only black-box testing.
3	Reliability or Security not performed in-depth, Focus on bugs, errors etc.	Reliability, Security & Robustness_checked during beta testing.
4	Long execution cycles maybe require.	Only a few weeks are required_for_the execution.
5	Critical issues can be identified by developers immediately.	Most of the issues or feedback is collecting from user will be implemented for the future versions of the product.
6	It focus on the product's quality before going to beta testing.	It focus on product's quality & ensures that the product is ready for real-time users.
7	It performed nearly the end of the software development.	It is a final test before shipping a product to the customers.